

ILES Deployment Guide: Render Backend + Vercel Frontend

This guide deploys the Internship Logging & Evaluation System (ILES) from GitHub.

Repository:

<https://github.com/holosion/ILES.git>

What Will Be Deployed

Important Files Added

Step 1: Push Latest Code to GitHub

From the project root:

```
cd C:\Users\akena\OneDrive\Desktop\groupc
git status
git add .
git commit -m "Prepare Render and Vercel deployment"
git push origin master
```

If Git says there is nothing to commit, the latest deployment files are already pushed.

Step 2: Deploy the Backend on Render

Render official Django guidance recommends PostgreSQL, WhiteNoise, a build script, and Gunicorn.

1. Open Render: <https://dashboard.render.com>
2. Sign in or create an account.
3. Click ****New +****.
4. Choose ****Blueprint****.
5. Connect GitHub if Render asks for permission.
6. Select the repository:

`holosion/ILES`

7. Render will read `render.yaml`.
8. Confirm it will create:
 - `iles-backend`
 - `iles-db`
9. Click ****Apply****.
10. Wait for the build to finish.

Expected backend URL format:

`https://iles-backend.onrender.com`

Your actual URL may include a different suffix if the name is already taken.

Step 3: Test the Render Backend

Open this in a browser after Render finishes:

```
https://YOUR_RENDER_BACKEND_URL/api/dashboard/
```

Example:

```
https://iles-backend.onrender.com/api/dashboard/
```

You should see JSON like:

```
{
  "total_students": 0,
  "total_reports": 0
}
```

If you see a Django error:

1. Open the Render service.
2. Click **Logs**.
3. Check whether migrations, static collection, or environment variables failed.

Step 4: Deploy the Frontend on Vercel

Vercel official Vite guidance uses normal Vite build output and VITE_ environment variables.

1. Open Vercel: <https://vercel.com/dashboard>
2. Sign in or create an account.
3. Click **Add New...**
4. Choose **Project**.
5. Import the GitHub repository:

```
holosion/ILES
```

6. Set **Root Directory** to:

```
frontend
```

7. Use these build settings:

8. Add this environment variable:

Example:

```
VITE_API_BASE_URL=https://iles-backend.onrender.com/api
```

9. Click **Deploy**.

Expected frontend URL format:

```
https://your-project-name.vercel.app
```

Step 5: Test the Full Live System

Open the Vercel URL.

Test this flow:

1. Create a company account.
2. Copy the demo verification code shown in the app.
3. Verify the account.
4. Add a student profile.
5. Generate a weekly report.
6. Log out.
7. Create a lecturer account.
8. Verify the lecturer account.
9. Open **Reports**.
10. Select a company.
11. Select a student.
12. Enter a mark from 1 to 100.
13. Confirm the system calculates the Makerere grade and final mark.

Step 6: If Login Does Not Work

Check these in order:

1. Is the Render backend awake?
 - Open https://YOUR_RENDER_BACKEND_URL/api/dashboard/.
2. Is Vercel using the right API URL?
 - Vercel Project > Settings > Environment Variables.
 - Confirm `VITE_API_BASE_URL=https://YOUR_RENDER_BACKEND_URL/api`.
3. Did you redeploy Vercel after changing environment variables?
 - Environment variable changes only affect new deployments.
4. Is CORS blocking the request?
 - The backend allows *.vercel.app domains and local development URLs.

Step 7: Future Updates

After changing code locally:

```
cd C:\Users\akena\OneDrive\Desktop\groupc
git add .
git commit -m "Describe your change"
git push origin master
```

Render and Vercel should automatically redeploy after the push if auto-deploy is enabled.

Commands for Local Development

Backend:

```
cd C:\Users\akena\OneDrive\Desktop\groupc\backend
python -m pip install -r requirements.txt
python manage.py migrate
python manage.py runserver 127.0.0.1:8000
```

Frontend:

```
cd C:\Users\akena\OneDrive\Desktop\groupc\frontend
npm install
npm run dev
```

Local URL:

<http://127.0.0.1:5173>

Official References

- Render Django deployment docs: [`https://render.com/docs/deploy-django`](https://render.com/docs/deploy-django)
- Render deploy concepts: [`https://render.com/docs/deploys/`](https://render.com/docs/deploys/)
- Vercel Vite docs: [`https://vercel.com/docs/frameworks/frontend/vite`](https://vercel.com/docs/frameworks/frontend/vite)
- Vercel environment variables docs: [`https://vercel.com/docs/projects/environment-variables`](https://vercel.com/docs/projects/environment-variables)